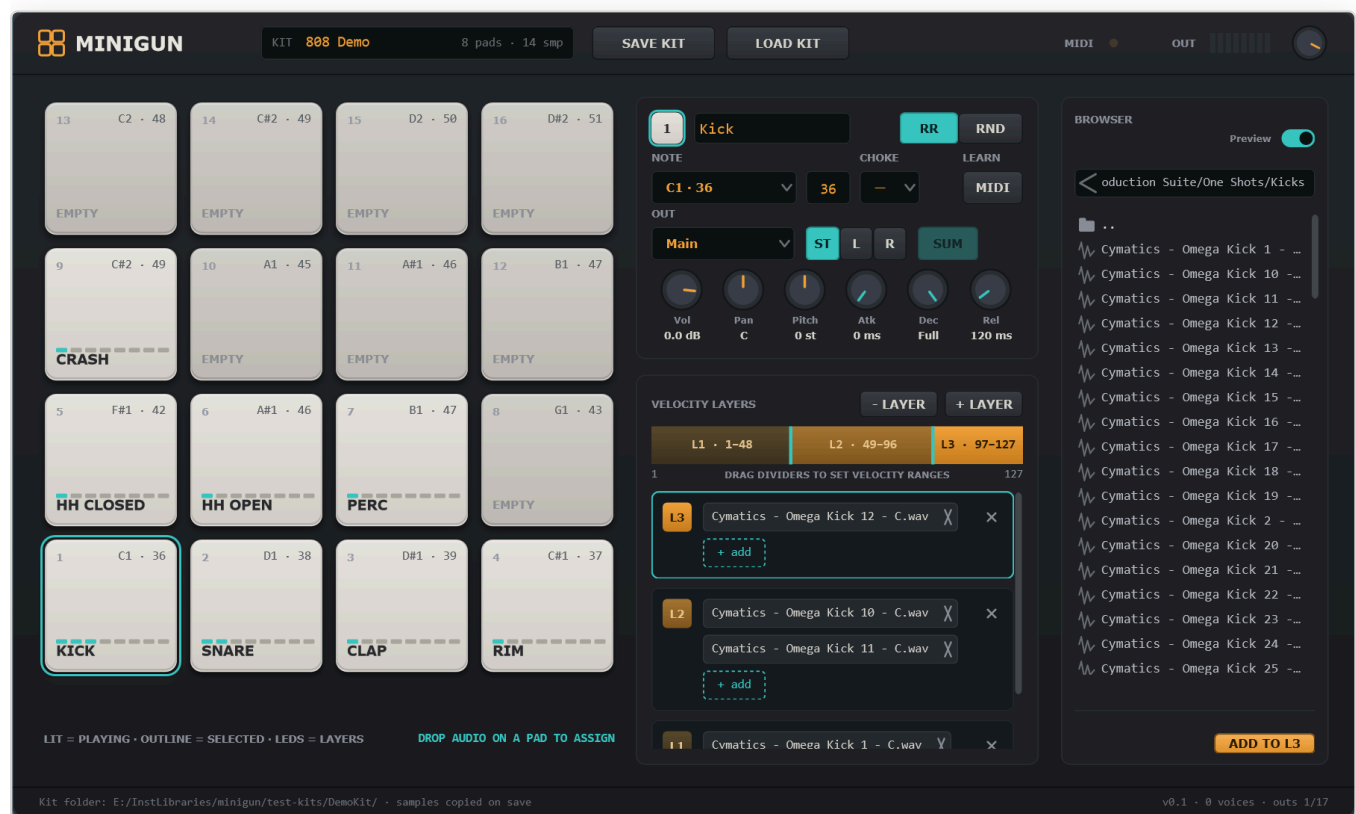


MINIGUN

# Minigun Drum Sampler

User Manual · VST3 & Standalone · version 0.4.0



16 pads · velocity layers · round-robin / random · portable kits · 17 stereo outputs  
Windows 10/11, 64-bit. Built with JUCE 8.

# 1. What Minigun is

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Minigun is a lightweight drum sampler in the spirit of hardware pad machines. Sixteen pads each hold one instrument: a MIDI note, up to eight velocity layers, and any number of samples per layer that alternate round-robin or at random. Every pad has its own volume, pan, pitch, envelope, choke group and output bus, so a whole kit can be mixed inside your DAW on separate tracks.

Kits are **portable**: when you save a kit, Minigun copies every sample into the kit folder and stores relative paths, so the folder can be moved to another project or another computer without broken links.

## Installation

- **VST3**: copy `Minigun.vst3` (the whole folder) to `C:\Program Files\Common Files\VST3\`, then rescan plug-ins in your DAW. Minigun appears as *VST3i: Minigun (Dallas Audio)*.
- **Standalone**: run `Minigun.exe`. Use *Options* → *Audio/MIDI settings* to pick an audio device and a MIDI input. Only the Main output is available in Standalone.

The Standalone version and a DAW that holds an exclusive ASIO driver cannot use the same audio device at the same time. If Standalone shows *0 voices* and no meter movement, close the DAW or choose another device.

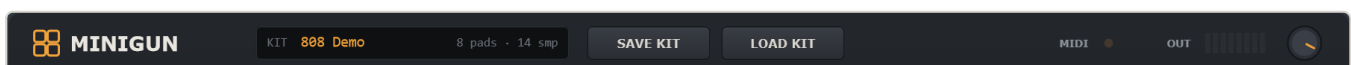
## 2. Interface overview



The fixed 1200 × 720 window. Everything is always visible; nothing is hidden in menus.

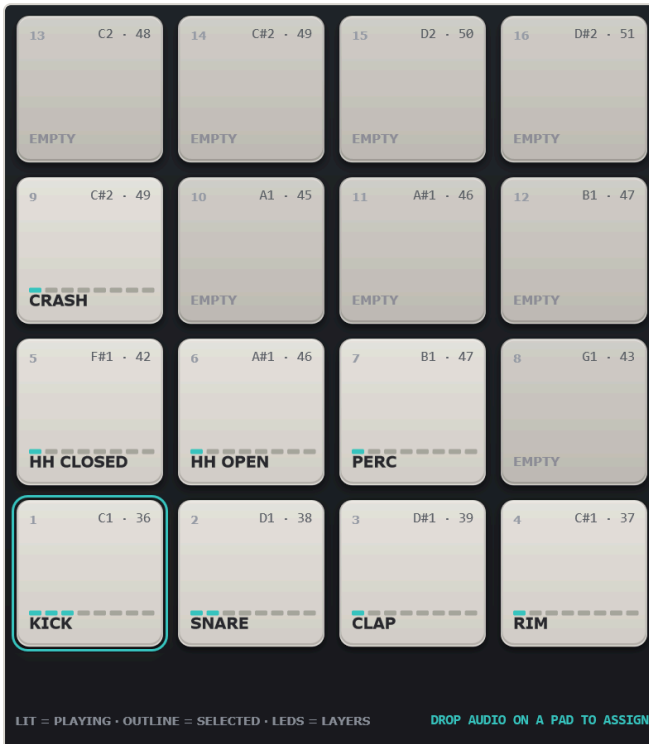
- 1 **Header** — kit name, Save / Load kit, Undo / Redo, MIDI indicator, output meter, master volume.
- 2 **Pads** — 16 pads in MPC order (pad 1 bottom-left, pad 13 top-left). Click to play, drop audio files to assign.
- 3 **Advanced strip** — under the pads: per-hit humanize (RND PITCH / RND VOL) and the kit-wide TRIG switch.
- 4 **Pad editor** — every setting of the selected pad: name, note, choke, output, mix and envelope knobs.
- 5 **Velocity layers** — the velocity map of the selected pad and the samples inside each layer.
- 6 **Browser** — a file browser with instant preview; drag files onto pads or layers.
- 7 **Footer** — kit folder, active voices, and how many output buses the host has enabled.

## 3. Header



| Control     | What it does                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KIT display | Shows the kit name, how many pads have samples and the total sample count. The name is taken from the folder you save into (a kit called <i>Untitled Kit</i> is renamed automatically on first save).                                                                                                                                                                                                                    |
| SAVE KIT    | Saves the kit. The first time it opens a folder picker (default location: <code>Documents\Minigun Kits</code> ). Choose or create a folder. Minigun writes <code>kit.json</code> there and <b>copies every sample</b> into a <code>samples\</code> sub-folder. If the kit already has a folder, it saves there directly without asking.                                                                                  |
| LOAD KIT    | Opens a folder picker. Select a folder that contains <code>kit.json</code> ; all pads, layers and settings are replaced by that kit.                                                                                                                                                                                                                                                                                     |
| UNDO / REDO | Step back and forward through every kit edit: knob moves, names, notes, layers, dropped samples, even a loaded kit. A continuous knob sweep or typing a name counts as one step. Keyboard: <code>Ctrl + Z</code> undo, <code>Ctrl + Y</code> or <code>Ctrl + Shift + Z</code> redo (the plug-in window must have keyboard focus: click on it first). Up to 200 steps; the history is cleared when a project is reopened. |
| MIDI        | Lights amber for a moment whenever a MIDI note-on arrives. Use it to confirm the DAW is routing MIDI to the plug-in.                                                                                                                                                                                                                                                                                                     |
| OUT         | Eight-segment stereo peak meter of the plug-in output (the loudest of all enabled buses). Teal segments are healthy, amber is hot.                                                                                                                                                                                                                                                                                       |
| Master knob | Master volume, $-60 \dots +6$ dB. This is the only DAW-automatable parameter; everything else belongs to the kit.                                                                                                                                                                                                                                                                                                        |

## 4. Pads



Pad 1 is selected (teal outline). Loaded pads show their name and one LED per velocity layer.

### What a pad shows

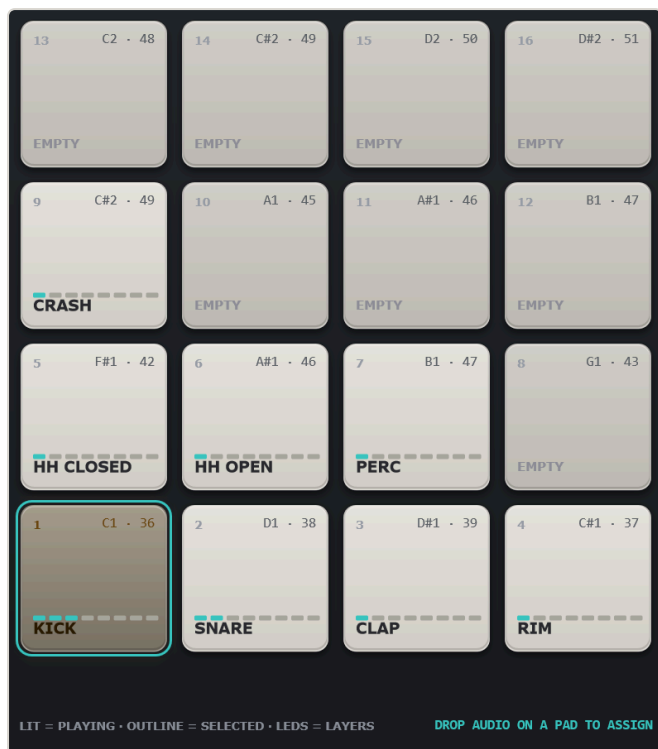
- **Number** (top-left) — pad 1...16 in MPC order.
- **Note** (top-right) — the MIDI note that triggers the pad, e.g. `C1 · 36`.
- **LED strip** — one lit LED per velocity layer (up to 8).
- **Name** — the pad name, or *EMPTY*.

### Pad states

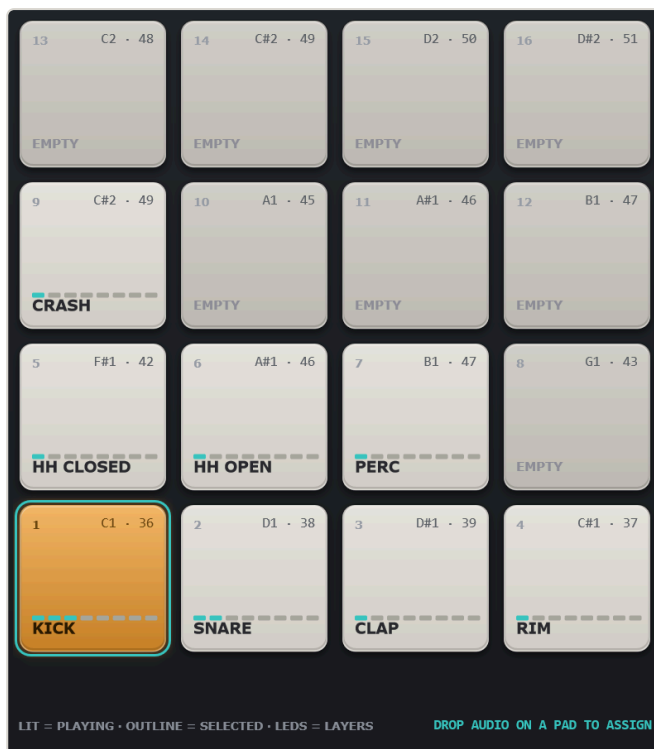
- **Empty** — darker rubber, no LEDs.
- **Loaded** — light rubber, LEDs show the layer count.
- **Selected** — teal outline; the editor and layer panel show this pad.
- **Playing** — the pad lights up in the colour of the velocity layer that played (see §6).

### Interactions

- **Click** a pad to select it and play it. The velocity follows the click height: top edge = 127, bottom edge = 1.
- **Drop audio files** from Explorer or from the Browser onto a pad. They join the pad's **active layer** (the layer selected in §6; for a pad that is not selected, its top layer). An empty pad gets its first layer (1–127) automatically. New layers are only created with **+ LAYER**.
- A pad is named automatically after its first sample when it has no name yet.
- **Move a pad onto another pad** — built the pad in the wrong slot? Press it and drag it onto the slot you wanted. The target pad turns amber and reads *REPLACE* (or *MOVE HERE* when it is empty). On release Minigun asks for confirmation: **Move** transfers the samples and every pad setting and **overwrites** the target, **Cancel** does nothing. The original pad is left empty. Notes do not travel: each slot keeps its own MIDI note. `Ctrl1 + Z` undoes the move.



A soft hit (layer 1) lights the Kick pad dark brown...



...a hard hit (layer 3) lights it bright amber. The colours match the layer bar in §6.

## 5. Pad editor

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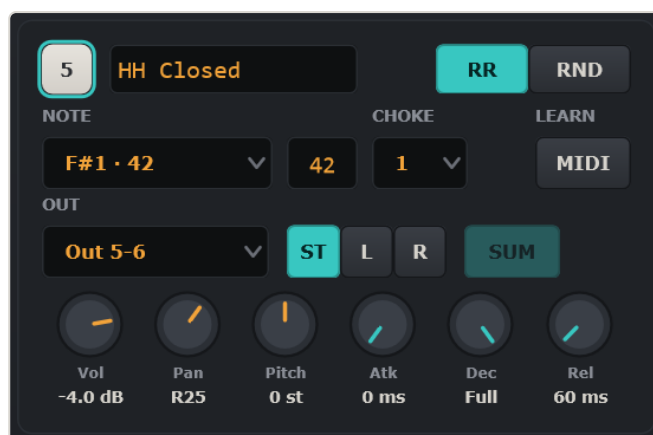
The editor always shows the selected pad (badge 1 = pad 1).

| Control      | What it does                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Badge + name | The pad number and an editable name. Click the field and type: the name is saved as you type and shown on the pad; <b>Enter</b> , <b>Esc</b> or a click anywhere else leaves the field.                                                                                                                                                                                                                                                                    |
| RR / RND     | How samples inside a layer alternate. <b>RR</b> (round-robin) plays them in order 1, 2, 3, 1...; <b>RND</b> picks one at random on every hit. Use several similar samples per layer for a natural, non-machine-gun feel.                                                                                                                                                                                                                                   |
| NOTE         | The MIDI note that triggers the pad. Pick it from the drop-down (all 128 notes, <b>C1 · 36</b> style) or type the number into the field next to it and press <b>Enter</b> . The mouse wheel over the drop-down steps by one semitone. Several pads may share a note: they all play together.                                                                                                                                                               |
| CHOKE        | Choke group, — or 1...8. Pads in the same group cut each other off: when one is hit, the others in that group are released immediately. Classic use: closed and open hi-hat in group 1, so the closed hat stops the open one.                                                                                                                                                                                                                              |
| LEARN / MIDI | MIDI learn. Click the button (it lights teal), then hit a key or pad on your controller: that note is assigned to the selected pad.                                                                                                                                                                                                                                                                                                                        |
| OUT          | Output bus of the pad: <b>Main</b> or one of the 16 stereo pairs <b>Out 1-2 ... Out 31-32</b> . See §9 for routing in the DAW.                                                                                                                                                                                                                                                                                                                             |
| ST / L / R   | Stereo or mono output. <b>ST</b> sends the pad to both channels of the pair with panning. <b>L</b> sends a mono signal to the first (odd) channel only, <b>R</b> to the second (even) channel. In mono mode the OUT list shows single channels ( <i>Main L, Out 3, ...</i> ) and Pan is ignored.                                                                                                                                                           |
| VEL          | Velocity → volume. <b>On</b> (default): the MIDI velocity scales the pad's level, so a soft hit is quieter (velocity 64 ≈ −6 dB, 32 ≈ −12 dB). <b>Off</b> : every hit plays at full level — the velocity still chooses the layer and the RR/RND sample, it just no longer changes the volume. Useful for 808-style one-shots and for kits whose layers are already recorded at the right level. <b>Ctrl</b> + click the button to set all 16 pads at once. |
| SUM          | Only for L/R modes. On: a stereo sample is summed to mono ((L+R)/2) before it goes to the single channel. Off: only the sample channel that matches the side is used. Mono samples are unaffected.                                                                                                                                                                                                                                                         |
| Vol          | Pad volume, −60 ... +12 dB.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Pan          | Stereo position, L100 ... C ... R100 (constant power). Not used in mono modes.                                                                                                                                                                                                                                                                                                                                                                             |
| Pitch        | Transposition by resampling, ±12 semitones in 0.1 steps. Positive values are shorter and brighter.                                                                                                                                                                                                                                                                                                                                                         |
| Atk          | Attack, 0 ... 500 ms: fade-in from the start of the sample. 0 ms = the natural transient.                                                                                                                                                                                                                                                                                                                                                                  |
| Dec          | Decay / hold length, 1 ... 5000 ms. Fully right = <b>Full</b> : the sample plays to its end. Shorter values cut the tail and start the release.                                                                                                                                                                                                                                                                                                            |
| Rel          | Release, 1 ... 2000 ms: the fade-out applied when the decay time ends, or when the pad is choked by another pad of its group.                                                                                                                                                                                                                                                                                                                              |

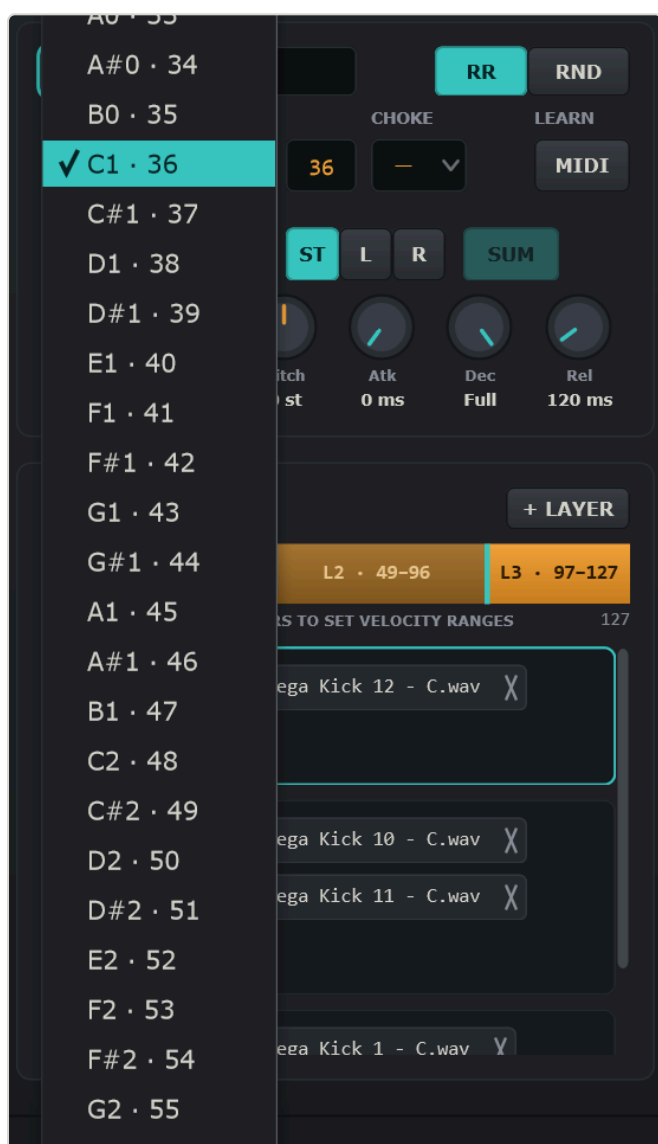




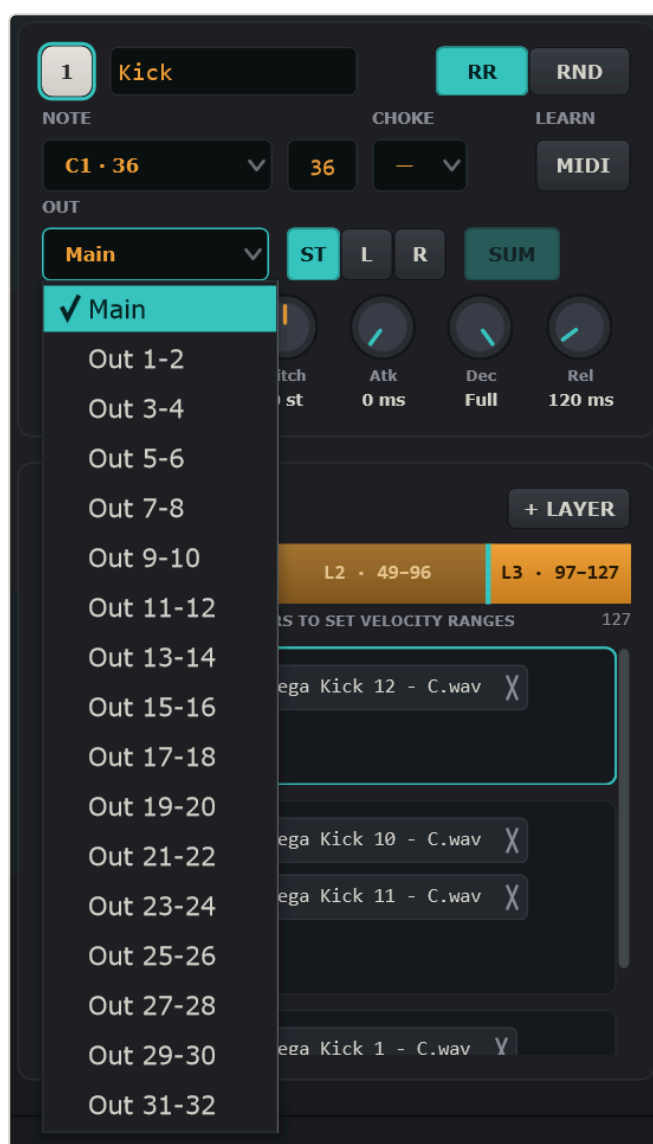
Snare pad routed to the stereo pair *Out 1-2*.



Closed hi-hat: choke group 1, panned right, *Out 5-6*, short 60 ms release.



The NOTE drop-down lists all 128 notes with their numbers.



The OUT drop-down: Main plus 16 stereo pairs.

Knob gestures

Every knob has a value field under it. **Click the field, type a value and press `Enter`** to set it exactly; the field accepts what it shows, so Pan takes `L40`, `C` or `R75` and Dec takes `Fu11`. **`Ctr1` + click a knob** (or double-click it) snaps it back to its default: 0.0 dB, C, 0 st, 0 ms attack, Full decay, 120 ms release, and OFF for the two RND knobs.

Advanced · Humanize



The advanced strip sits under the pads: TRIG, RND PITCH and RND VOL.

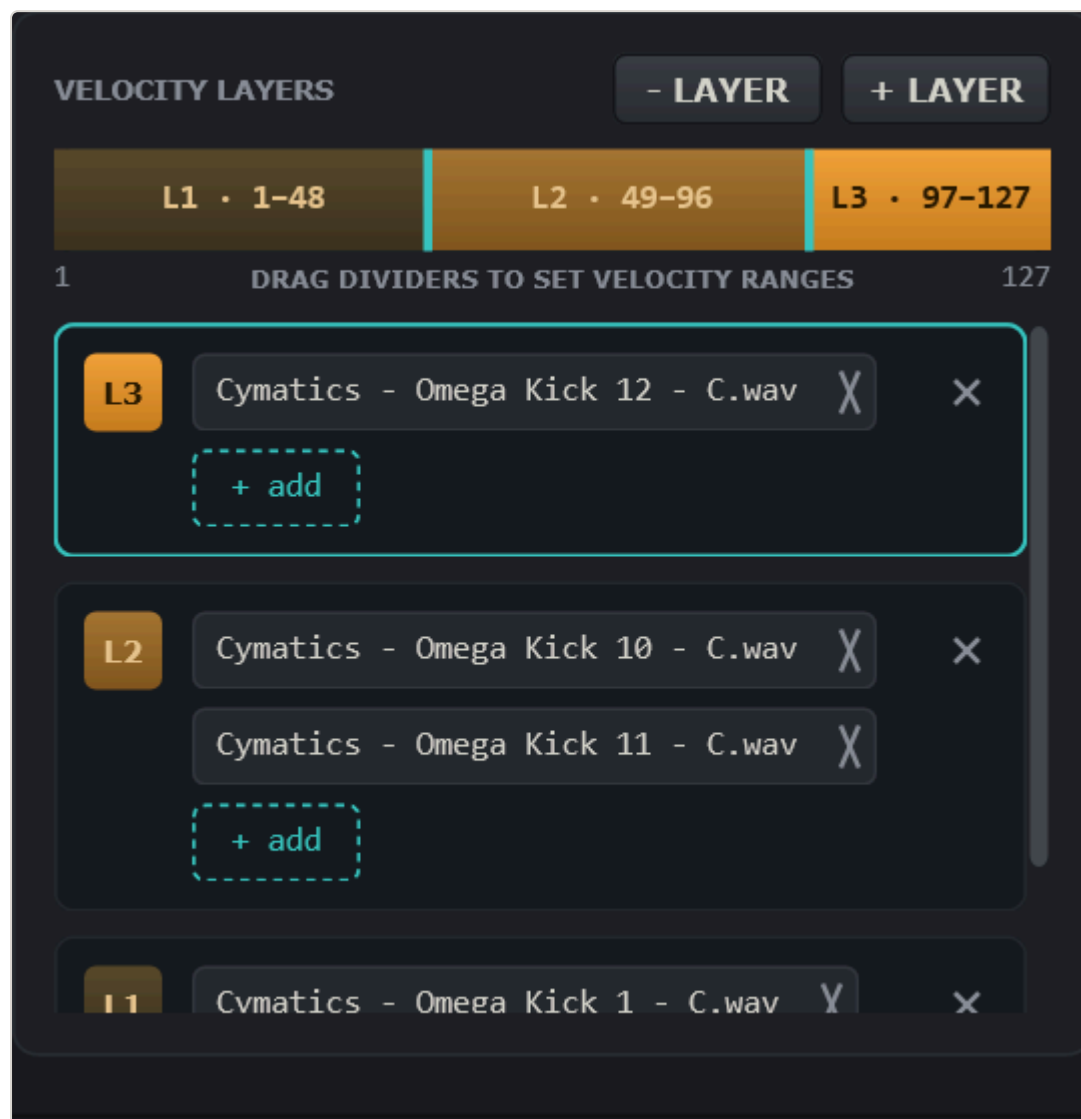
When the same sample is fired again and again — blast beats, double kick, a snare roll — every hit is bit-for-bit identical, they comb with each other and the part starts to sound like a machine-gun instead of a drummer. Round-robin samples fix that when you have them; humanize fixes it when you do not.

| Control   | What it does                                                                                                                                                                                                                                                                                                             |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RND PITCH | Per-pad random detune, <b>0 ... 50 cents</b> either way, drawn fresh on every hit. 8–12 c is enough to break the combing while staying inaudible as detune; past ~25 c the drum starts to sound out of tune. This is the knob that kills the machine-gun effect.                                                         |
| RND VOL   | Per-pad random level, <b>0 ... 6 dB</b> either way, fresh on every hit. 0.5–1.5 dB makes a repeated hit breathe; above 2 dB the groove begins to wobble.                                                                                                                                                                 |
| TRIG      | Trigger mode for the whole kit: switch it on when the kit is played from a programmed or drum-replaced MIDI track rather than by hand. Every pad then humanizes by <b>10 cents and 1.0 dB</b> and the two knobs are locked (they show the amount being applied). Switch it off and each pad goes back to its own values. |

**Timing is never touched.** Minigun starts every note exactly on the sample it arrives on, so humanize adds no shuffle, no flam and no latency — the groove stays as tight as it was programmed.

Example — a 16th-note double-kick pattern on one sample: set RND PITCH to 10 c and RND VOL to 1.0 dB (or just press TRIG). The individual hits stop cancelling each other and the run reads as a drummer's foot instead of one sample looped.

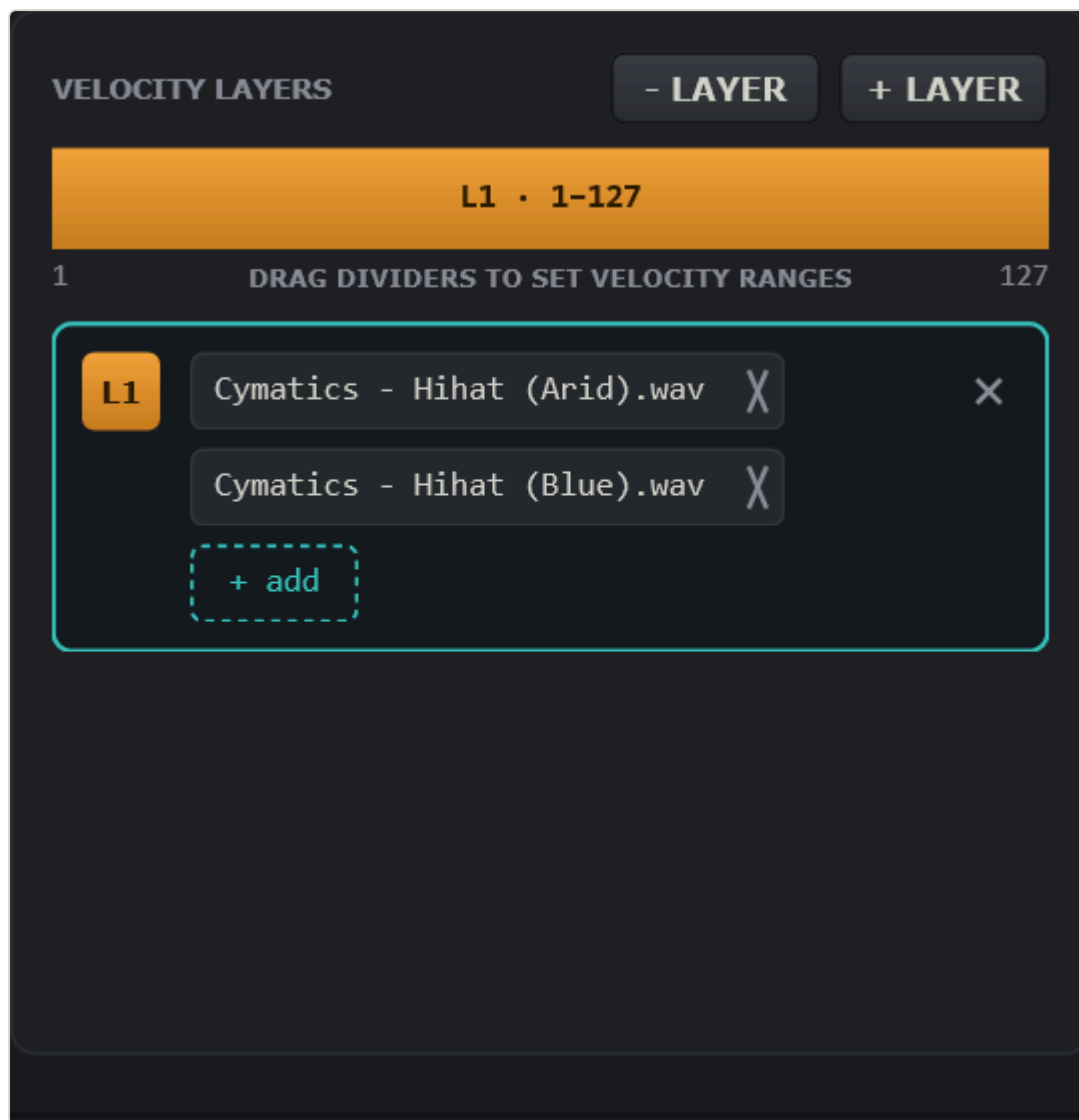
## 6. Velocity layers



Three layers on the Kick pad. Layer 2 holds two samples that alternate round-robin.

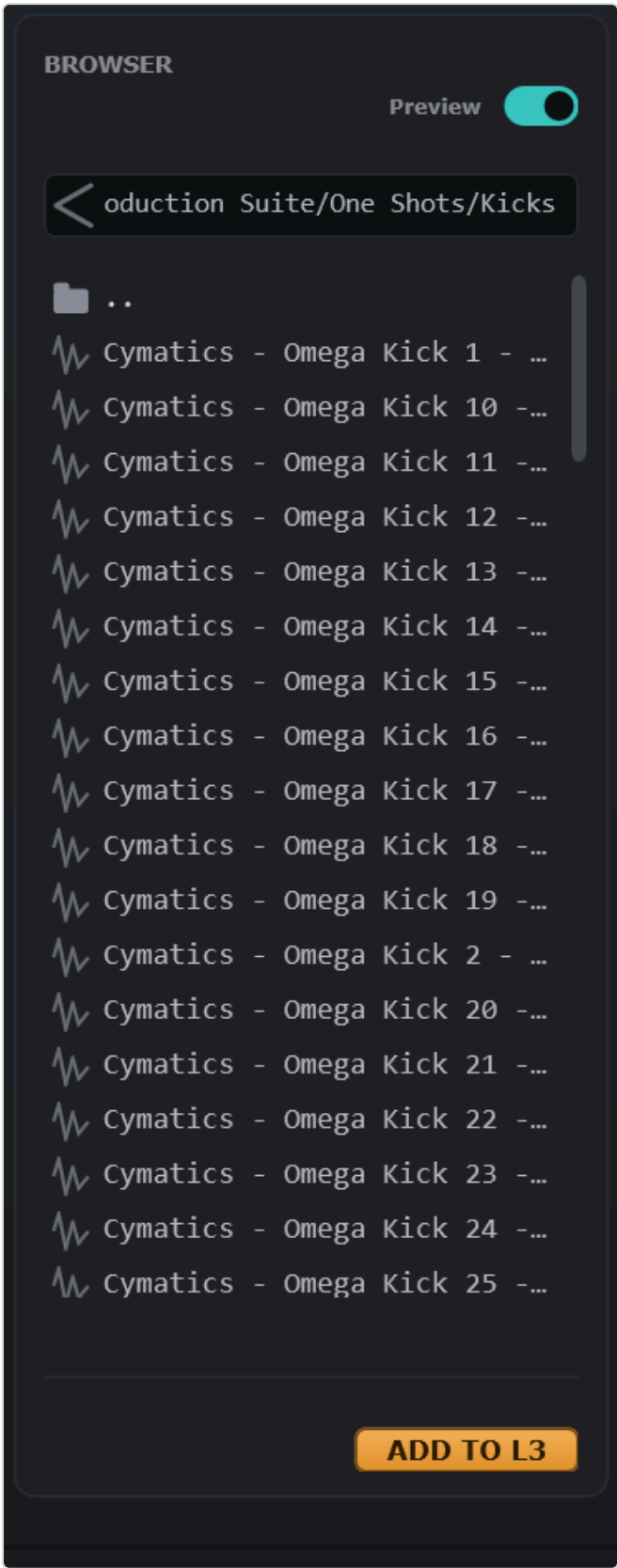
A velocity layer is a velocity range (1–127) with its own list of samples. When a note arrives, Minigun finds the layer whose range contains the velocity, then picks one of its samples by the pad's RR/RND rule. Layers are contiguous and never overlap; the panel keeps them valid for you.

| Control           | What it does                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range bar         | One coloured segment per layer, from dark (soft) to bright amber (hard). <b>Drag the teal dividers</b> to move the boundaries. <b>Click inside a segment</b> to audition that layer: the pad plays at the clicked velocity, so the layer and the RR/RND rule apply exactly as with a MIDI note.                                                                                      |
| + LAYER           | Adds a layer on top: the current top layer is split in half and the new (empty) layer takes the upper part. Up to 8 layers.                                                                                                                                                                                                                                                          |
| Layer rows        | One row per layer, top layer first. The coloured badge ( <i>L1</i> , <i>L2</i> ...) matches the bar. Click a row to make it the <b>selected layer</b> (teal frame): the Browser's <i>ADD TO Lx</i> button and double-clicks add files to it.                                                                                                                                         |
| Sample chip       | One chip per sample. <b>Click</b> the name to preview the sample. <b>Click x</b> to remove it from the layer (the file on disk is not touched). <b>Drag</b> a chip to reorder it inside its layer or to move it to another layer row / bar segment: a teal caret shows the exact slot where it will land, and holding <b>Ctrl</b> on release copies the sample instead of moving it. |
| + add             | Opens a file dialog (multi-select) and adds the chosen files to that layer.                                                                                                                                                                                                                                                                                                          |
| × (row) / – LAYER | Deletes a layer: the × in the top-right corner of a row removes that layer; – <b>LAYER</b> in the panel header removes the selected layer. Right-click on a row offers the same <i>Delete layer</i> . The removed range goes to the neighbour below (or above for the lowest layer); samples on disk are untouched.                                                                  |
| Drop target       | Files dragged from Explorer or the Browser can be dropped on a row or on a bar segment (that layer) or anywhere else in the panel (the selected layer). The target is highlighted with a dashed teal frame while you drag.                                                                                                                                                           |



A single full-range layer with two hi-hat samples: every hit alternates between them.

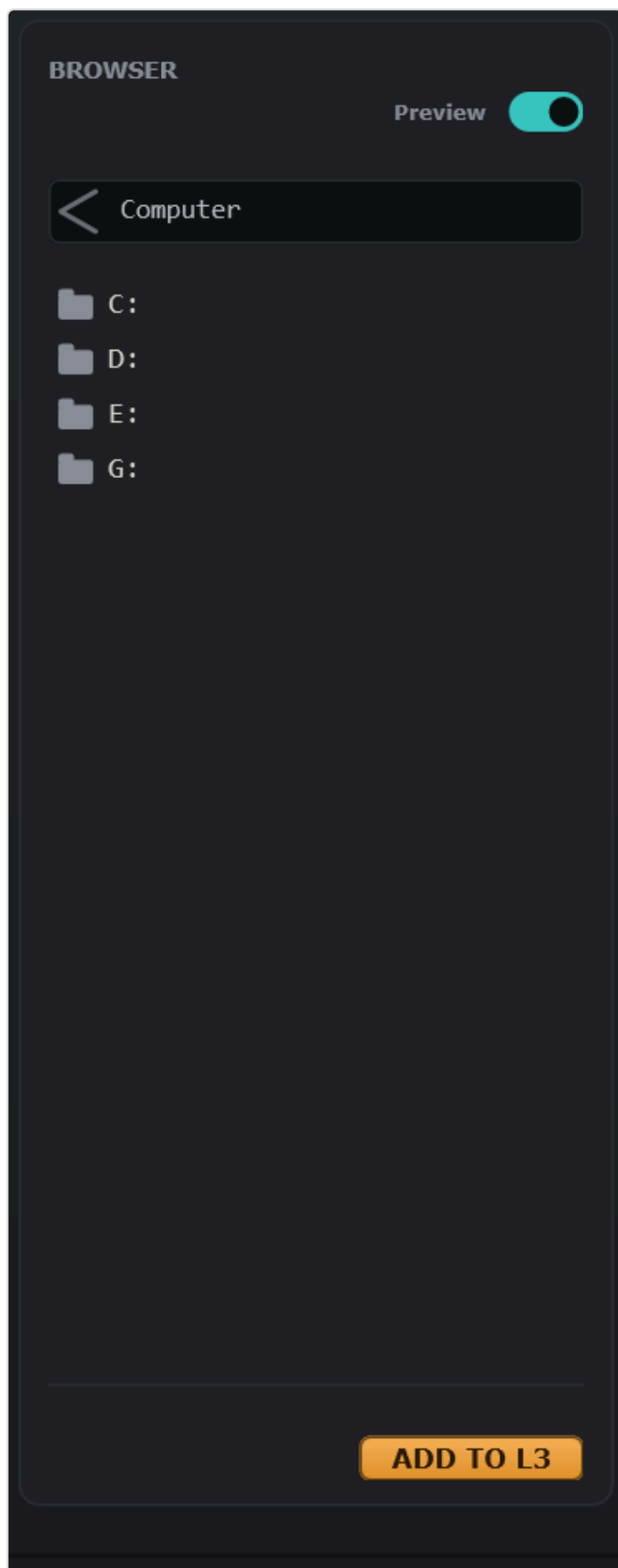
# 7. Browser



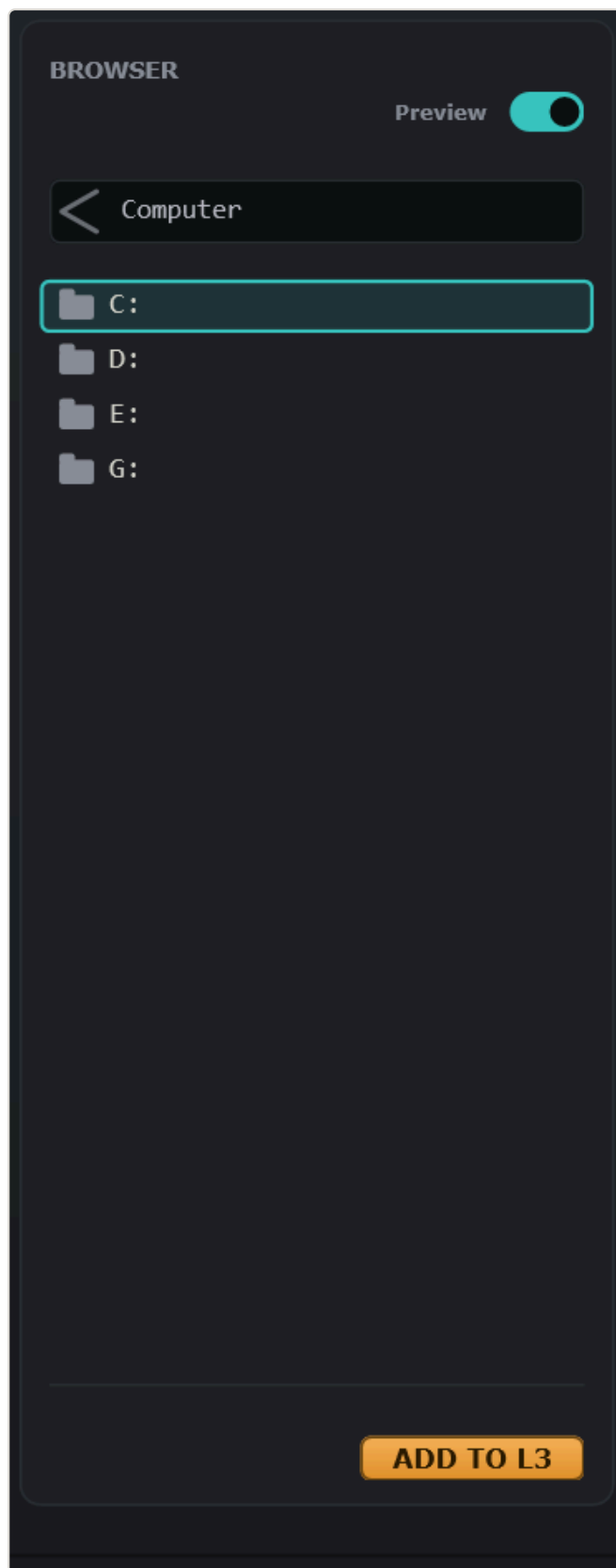
Audio files show a waveform icon; the playing file shows a play triangle. The footer shows the waveform and format of the selected file.

| Control         | What it does                                                                                                                                                                            |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preview         | When on, a single click on a file plays it through the plug-in's Main output. Turn it off to browse silently.                                                                           |
| < (arrow)       | Go up one folder. From the root of a drive it opens the <i>Computer</i> level with all drives.                                                                                          |
| Path field      | Shows the current folder. Click it, type a path to a folder or to an audio file and press <b>Enter</b> . A file path opens its folder and selects the file. <b>Esc</b> reverts.         |
| Folder row      | Single click frames the folder; double-click or <b>Enter</b> opens it.                                                                                                                  |
| File row        | Single click selects and previews. Double-click or <b>Enter</b> adds the file to the selected layer of the selected pad. Drag a row onto a pad or onto the layer panel to add it there. |
| Waveform + info | Overview of the selected file with sample rate, bit depth and length.                                                                                                                   |
| ADD TO Lx       | Adds the selected file to layer x of the selected pad (x = the layer selected in §6).                                                                                                   |

Supported formats: WAV, AIFF, FLAC, OGG, MP3, WMA. The last folder is remembered between sessions.



The *Computer* level lists all drives.



A clicked folder gets a teal frame; double-click to enter.

## Navigation shortcuts

| Action                                                                                                                                                                                          | Result                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Backspace</b>                                                                                                                                                                                | Up one level.                                                                      |
| <b>Mouse Back / Forward buttons</b>                                                                                                                                                             | Back / forward through the folder history (while the pointer is over the browser). |
| <b>Alt</b> +  / <b>Alt</b> +  | Back / forward through the history.                                                |
| <b>Enter</b>                                                                                                                                                                                    | Open the framed folder or add the selected file.                                   |
|  /                            | Move through the list.                                                             |



## 8. Kits: saving, loading, moving

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A kit is a folder. **SAVE KIT** creates this structure:

|             |                                                |
|-------------|------------------------------------------------|
| My Kit\     |                                                |
| kit.json    | all pad settings, layers and sample references |
| samples\    |                                                |
| kick_01.wav | every sample used by the kit, copied here      |
| kick_02.wav |                                                |
| ...         |                                                |

- Sample paths inside `kit.json` are relative to the kit folder, so the folder can be zipped, moved to another disk or another computer, and loaded with **LOAD KIT** without any missing files.
- Files that are already inside `samples\` are not copied again. If two different files share a name, the second gets a `_2` suffix.
- Removing a chip never deletes a file from disk.
- In a DAW the complete kit is also stored in the project, so a project reopens exactly as it was saved even if you never pressed SAVE KIT. Save a kit when you want to reuse it in another project.

If a sample cannot be found when a kit or project is loaded, the pad keeps its settings but that sample is silent. Add the file again from the Browser or replace it via + *add*.

## 9. Routing pads to separate DAW tracks

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Minigun exposes **17 stereo outputs**: Main and Out 1-2 ... Out 31-32 (34 channels). Each pad chooses its bus with the OUT drop-down, and can send a mono signal to a single channel with the L / R modes.

### REAPER example

1. Insert Minigun on a track. The FX window shows a button like *2/34 out*.
2. Set the track's channel count (track routing button) to as many channels as you need, e.g. 8 for Main + Out 1-2 + Out 3-4 + Out 5-6. REAPER enables exactly that many plug-in buses; the footer shows *outs 4/17*.
3. In Minigun set the Snare pad to *Out 1-2*, the Clap to *Out 3-4*, the hats to *Out 5-6*.
4. Create receive tracks: on a new track add a receive from the Minigun track with source channels 3/4 for the snare, 5/6 for the clap, 7/8 for the hats. Or use *Build multichannel routing for this plug-in* in the plug-in I/O window.

Other DAWs work the same way: enable the plug-in's aux outputs and route each pair to its own track. A pad set to an output the host has not enabled falls back to Main, so nothing ever goes missing.

### Mono outputs

To send a pad to one channel only, choose the pair in OUT and set **L** (odd channel) or **R** (even channel). With **SUM** on, a stereo sample is folded to mono; with SUM off, only the matching side of the sample is used.

## 10. Step-by-step examples

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### A kick with three velocity layers

1. Click pad 1. In the Browser go to your kick samples.
2. Double-click the softest sample: it becomes layer 1 (1–127).
3. Press + **LAYER**: layer 2 appears above (65–127). Double-click a medium sample to add it there, then a second medium sample for round-robin.
4. Press + **LAYER** again and add the hardest sample to layer 3.
5. Drag the dividers so the ranges feel right, e.g. 1–48, 49–96, 97–127. Click inside each segment to audition it.

### Closed and open hi-hat that choke each other

1. Drop a closed hat on pad 5 and an open hat on pad 6.
2. Set **CHOKE** to 1 on both pads.
3. Give the closed hat a short **Rel** (about 60 ms) so it cuts the open hat cleanly.

### A snare on its own mixer track

1. Select the snare pad, set **OUT** to *Out 1-2*.
2. In the DAW enable the plug-in's second output pair and route channels 3/4 to a new track.

### A mono percussion hit on channel 5

1. Select the pad, set **OUT** to *Out 5-6*, press **L**: the list now reads *Out 5*.
2. Leave **SUM** on if the sample is stereo and you want both sides folded in.

### Assign a controller pad

1. Select the pad, press **MIDI** (LEARN).
2. Hit the pad on your controller. The note is shown in NOTE and on the pad.

## 11. Shortcuts and gestures

| Where                    | Gesture                                                   | Result                                                                       |
|--------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| Pad                      | Click (top / bottom)                                      | Play loud / soft, select the pad                                             |
| Pad                      | Drop files                                                | Add the files to the active layer                                            |
| Pad                      | Drag onto another pad                                     | Move the pad there after confirming (overwrites it, empties the original)    |
| Range bar                | Click segment                                             | Audition that layer (RR/RND applies)                                         |
| Range bar                | Drag divider                                              | Change velocity ranges                                                       |
| Layer row                | × / right-click, or – LAYER                               | Delete layer                                                                 |
| Chip                     | Click / click ×                                           | Preview / remove sample                                                      |
| Chip                     | Drag (hold <b>Ctrl</b> to copy)                           | Reorder inside the layer or move to another layer                            |
| VEL button               | <b>Ctrl</b> + click                                       | Set velocity → volume for all 16 pads                                        |
| Any knob                 | <b>Ctrl</b> + click or double-click                       | Back to the default value                                                    |
| Value field under a knob | Type a value + <b>Enter</b>                               | Set the knob exactly ( <b>L40</b> , <b>Full</b> , <b>OFF</b> are understood) |
| NOTE drop-down           | Mouse wheel                                               | ±1 semitone                                                                  |
| Browser                  | <b>Backspace</b>                                          | Up one folder                                                                |
| Browser                  | Mouse Back / Forward,<br><b>Alt</b> + <b>←</b> / <b>→</b> | History back / forward                                                       |
| Browser                  | Double-click / <b>Enter</b>                               | Open folder or add file to the selected layer                                |
| Path field               | Type path + <b>Enter</b>                                  | Jump to folder or file                                                       |
| Anywhere                 | <b>Ctrl</b> + <b>Z</b> / <b>Ctrl</b> + <b>Y</b>           | Undo / redo the last kit edit                                                |

## 12. Footer and troubleshooting

Kit folder: E:/InstLibraries/minigun/test-kits/DemoKit/ · samples copied on save

v0.1 · 0 voices · outs 1/17

- **Kit folder** — where the kit was saved or loaded from; (*unsaved*) until the first SAVE KIT.
- **voices** — samples currently sounding (32 maximum; the oldest is stolen when exceeded).
- **outs N/17** — how many output buses the host has enabled. In Standalone this is 1/17; in a DAW it grows with the track's channel count.

| Symptom                                                       | Check                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No sound in Standalone                                        | Options → Audio/MIDI settings: pick an output device. Another program (a DAW with ASIO) may hold the device exclusively.                                                                                                                                                |
| Pads light but a pad is silent                                | Its samples may be missing (moved or renamed on disk). Re-add them; then SAVE KIT so they are copied into the kit folder.                                                                                                                                               |
| Pad routed to Out 5-6 comes out of Main                       | The host has not enabled that bus. Increase the track channel count or enable the plug-in's outputs; the footer <i>outs</i> value shows how many are active.                                                                                                            |
| Plug-in not listed in the DAW                                 | Make sure <code>Minigun.vst3</code> is in <code>C:\Program Files\Common Files\VST3</code> and rescan plug-ins.                                                                                                                                                          |
| A pad sounds a hair late or phasey against another instrument | Minigun has been sample-accurate since v0.4.0 (hits used to snap to the audio-buffer boundary). If something still drifts, it is the source track's plug-in latency: a MIDI send does not carry PDC compensation, so the notes leave the sending track already shifted. |
| Wrong pad plays from the keyboard                             | Check NOTE on each pad; several pads may share a note. Use LEARN to assign quickly.                                                                                                                                                                                     |